

Ensemble Methods

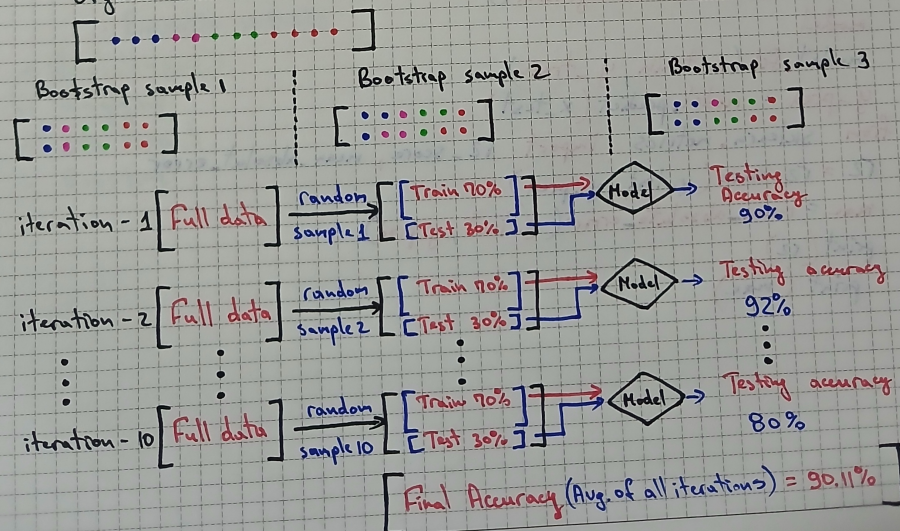
- ML technique
- Bir nechta kuchsiz modellarni birlashtirish orqali modelni rivojlantirish
- Asosiy logika: Kuchli kuchsiz algoritmlar kuchli model hosil qiladi
 - Nima uchun ishlatamiz:
 - Reduce variance $\rightarrow$ handle overfitting (e.g., bagging)
 - Reduce bias $\rightarrow$ improve weak learners (e.g., boosting)
 - Improve accuracy $\rightarrow$ aggregate multiple perspectives
 - Increase robustness $\rightarrow$ less sensitive to noise

low variance $\rightarrow$ kam bog'liqlik (low correlated) high variance $\rightarrow$ ko'p bog'liqlik (high correlated)
 bias $\rightarrow$ tarafdorlik (Yani: Target qiymat 0 $\rightarrow$ 30ta, 1 $\rightarrow$ 10 ta shunda balanced qilish kerak)
 Increase robustness $\rightarrow$ ma'lumotlar kopayganda model ishlashini ta'minlash (stabil)

Ensemble Methods: Turlari

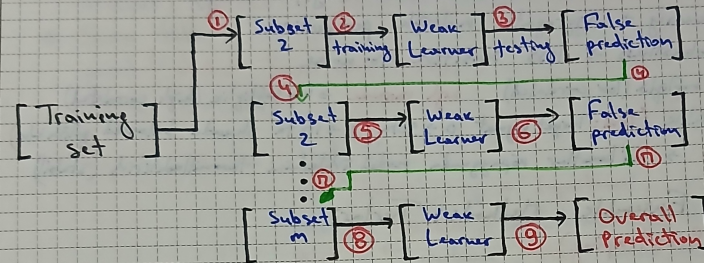
1. Bagging (Bootstrapping)
2. Boosting
3. Stacking
4. Voting

- Bagging (Bootstrapping) | Bootstrapping $\Rightarrow$ datasetni turli qimmlarga ajratish
 - Train multiple
 - Result: Voting / avg
 - Algorithm: Random Forest
 - Bir xil len, tushib qolish va qaytalanish normal



• Boosting:

- Train setna-ket
- Har bir model oldingisini to'g'irlaydi
- Yakuniy model = sum (kuchsiz)
- Algorithm: **GBM, Adaboost, XGB**
- Full dataset
- The Process of Boosting



Ensemble Methods: (Amaliy) ↓

Ensemble Methods

Bagging (Random Forest)

```

from sklearn.model_selection import train_test_split
X = df.drop('x6', axis=1)
y = df['x6']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=3)
from sklearn.ensemble import RandomForestRegressor
model = RandomForestRegressor()
model
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
from sklearn.metrics import r2_score, mean_absolute_error
r2 = r2_score(y_test, y_pred)
mae = mean_absolute_error(y_test, y_pred)
print(r2)
print(mae)

```

```

# Boosting (XGBoost)
# pip install xgboost
import xgboost as xgb
model = xgb.XGBRegressor(n_estimators=100, learning_rate=0.1, max_depth=3,
                          sub_sample=1, colsample_bytree=0.8, random_state=42)
model
model.fit(x_train, y_train)
y_pred = model.predict(x_test)
from sklearn.metrics import r2_score, mean_absolute_error
r2 = r2_score(y_test, y_pred)
mae = mean_absolute_error(y_test, y_pred)
print(r2)
print(mae)

```

Nimalar örgaydik

• SML

- Turtli xil algoritmlar:

- Ridge
- Lasso
- xGBoost

- Feature Engineering

- Yangi featuralar yaratish
- Feature transform
- Feature selection
- Ensemble methods

→ (Bootstrapping)

Bagging

- 2ta Project
- Boosting

• Python

- Built-in functions
- Classes
- Files